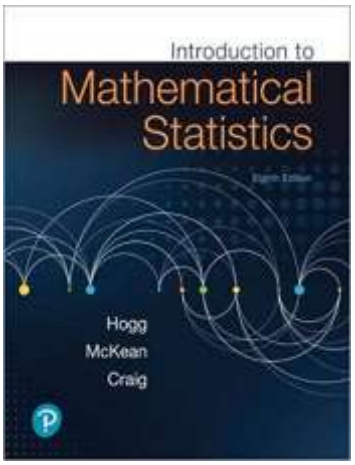
STAT 630 **Fall 2020**

MATHEMATICAL STATISTICS I

In the Fall semester of 2020, the University offers classes online. Thus, we'll have regular real-time lectures and real-time office hours via Zoom. Here is the link for all our classes: https://american.zoom.us/j/93234782658.
I will also recorded and publish our lectures for you to review. Quizzes and exams will be take-home.

Lectures	Monday & Thursday 6:55 – 8:20 pm	**Instructor**	Michael Baron
Room	Online via Zoom: https://american.zoom.us/j/93234782658	Office	DMTI 106-D
Course web site	http://fs2.american.edu/baron/www/630/	Phone	202-885-3130
Office hours	Proposed: Sunday & Wednesday 8:00 – 9:00 pm via Zoom: https://american.zoom.us/j/98611926068	Email	baron@american.edu



Textbook Introduction to Mathematical Statistics, 8th edition,
by R. V. Hogg, J. W. McKean, and A. Craig. Pearson, 2018
ISBN-10: 0134686993, ISBN-13: 978-0134686998
This book will be used for the next course STAT-631 as well!
Course web site - http://fs2.american.edu/baron/www/630/

Course plan:

1. Introduction. Foundations of Probability (1.1 – 1.4).
2. Random variables and their distributions. Moments. Transformations (1.5 – 1.9).
3. Probability inequalities (1.10).
4. Random vectors and multivariate distributions (2.1 – 2.8).
5. Discrete distributions: Binomial, Geometric, Negative Binomial, Multinomial, Hypergeometric, Poisson (3.1 – 3.2).
6. Continuous distributions: Normal, Gamma, t, F, Multivariate Normal (3.3 – 3.7).
7. Sampling distributions (4.1).
8. Statistical inference: estimation, confidence intervals, hypothesis testing (4.1 – 4.7).
9. Generation of random numbers. Monte Carlo methods and Bootstrap (4.8 – 4.9), if time permits.

Preparation… Your success in this course is based on good **working Calculus skills**. We'll routinely use derivatives, integrals, series, and limits. This includes functions of several variables, partial derivatives, and multiple integrals. Naturally, any statistic $S(X_1,\ldots,X_n)$, any quantity calculated from data, is a function of a random sample $X_1, \ldots, X_n$. So, it is a function of n variables. We'll often use Calculus to optimize it, finding either minimum or maximum, as in methods of least squares or maximum likelihood. There is no reason to use sub-optimal statistical methods, if we can find the very best solution!

So, a good **Calculus review** during the summer is a good idea. There are many sources available, and of course Calculus hasn't changed for the last 3 centuries! Your Calculus textbook and your notes should be the best sources because you are most familiar with them. To point the most relevant topics, let's look at these publicly available MIT lecture notes. For STAT-630/631 Mathematical Statistics, I would use these notes to review: Chap. 2; sections 3.1-3.4; 4.1, 4.3; 5.1-5.7; 6.2, 6.4; 7.1-7.2, 7.5; 8.1, 8.4; chap. 10; 13.2-13.3, 13.6-13.7; 14.1-14.2. These notes are also supplied by 18 videos.

Course requirements

The final grade will be based on weekly homework, short but regular weekly quizzes, a midterm test, and a final test.

Grading:

Weekly homework	15%	Homework will be assigned weekly. A steady effort to work out all the assigned problems is essential for learning theoretical principles and practical techniques of mathematical statistics and for the successful performance in this course. Homework solutions will be posted on Blackboard.
Weekly quizzes	25%	Short quizzes at the end of each week. Each quiz covers the material of the preceding week and the latest homework. Assigned every Friday via Blackboard, and they are due before Sunday 11:59 pm. Once you begin a quiz, you will have 45 minutes to complete and submit it.
Midterm test	25%	Each test covers several chapters of the material. The time is limited (2 hrs 30 min for the Midterm; 3 hrs 30 min for the Final), which means you have to be well prepared for each test. Taken on Blackboard. It is taken at home, but of course, serious preparation is essential for getting a good grade.
Final test	35%	

93 – 100 % = A	90 – 93 % = A-	87 – 90 % = B+	83 – 87 % = B
80 – 83 % = B-	77 – 80 = C+	73 – 77 % = C	70 – 73 % = C-

Software: During the course, we'll occasionally use **R**, including classroom demonstrations and examples. For all computer assignments, use the language of your choice. Advanced programming skills and advanced computer knowledge are *not* required. To use **R**, install it from https://cran.r-project.org/, free of charge. The same site has R manuals and other resources.

Learning outcomes: On successful completion of this course, students will be able to:

1. State and solve a given practical problem in mathematical terms, introducing relevant random variables and their distributions.
2. Calculate conditional and unconditional probabilities from the information given in word problems.
3. Propose a distribution model suitable for the described situation, calculate associated probabilities, expected values, variances, and moment generating functions.
4. Use mathematical tools such as sums and integrals to compute probabilities and moments that involve random variables or random vectors.
5. Derive new probability distributions that are related to the given ones.
6. Understand and apply standard methods of statistical estimation and testing.
7. Derive methods of generating random variables with given distributions and use them in Monte Carlo studies.
8. Use modern bootstrap methods for performance evaluation, confidence estimation, and hypothesis testing (if time permits).

Support Services

A wide range of services is available to support you in your efforts to meet the course requirements.

Mathematics & Statistics Tutoring Lab is now working online. Tutors should be able to help you with Calculus, Algebra, and basic Statistics, may be statistical software, but you should not count on getting homework solutions for graduate level Statistics courses! The Math & Stats Lab will be open Monday to Thursday from 11:00am-9:00pm and on Sundays from 3:00pm-9:00pm, offering both one-on-one and drop-in tutoring. See the schedule of one-to-one and drop-in hours.

To schedule a one-on-one tutoring session:

1. Visit american.mywconline.net, select the ONLINE Quant Support Schedule, and login
2. Search for your class using the drop down menu at the top of the page
3. Select a starting time with any tutor available and enter all the information in the pop-up window
4. Create the appointment and look out for emails from your tutor.

For questions, email jj8327a@american.edu or quantsupport@american.edu or visit http://www.american.edu/cas/mathstat/tutoring.cfm

CTRL Connect – software support with R (ctrl@american.edu, x2117)

Counseling Center (x3500) offers counseling and consultations regarding personal concerns, self-help information, and connections to off-campus mental health resources.

Academic Support and Access Center (x3360) offers study skills workshops, individual instruction, tutor referrals, Supplemental Instruction, writing support, and technical and practical support and assistance with accommodations for students with physical, medical, or psychological disabilities.

Religious Holidays. Students may receive accommodation in the course for the observance of a religious and/or cultural holiday. The student should notify the professor as soon as possible should such a need exist. More information about accommodations for religious and/or cultural holidays can be found at www.american.edu/ocl/kay/request-for-religious-accommodation.cfm.

Emergency Preparedness. In the event of an emergency, students should refer to the AU Web site (http://www.american.edu/emergency) and the AU information line at (202) 885-1100 for general university-wide information. In case of a prolonged closure of the University, I send updates to you by email and will post all announcements on the course web site.

Prerequisites: Calculus II and ability to take partial derivatives and multiple integrals.